CALCITRIOL

Hypercalcaemia

Active form — no renal activation

CHEMICAL NAME (1R,3S,5Z)-5-[(2E)-2-[(1R,3aS,7aR)-1-[(2R)-6-hydroxy-6-methylheptan-2-yl]-7a-methyl-octahydroinden-4-ylidene]ethylidene]-4-methylidenecyclohexane-1,3-diol (1,25-dihydroxycolecalciferol)

CLASS Active vitamin D (1,25-dihydroxyvitamin D3)

TRADE NAMES Rocaltrol · Calcijex (IV) · Silkis (topical)

THERAPEUTIC USES / INDICATIONS

- Renal osteodystrophy / secondary hyperparathyroidism in CKD.
- Hypoparathyroidism; hypocalcaemia.
- Vitamin D-resistant (hypophosphataemic) rickets.
- Postmenopausal osteoporosis (some countries); psoriasis (topical).

ADMINISTRATION

- **PO** capsules; **IV** in dialysis patients.
- Separate from bile-acid sequestrants.

DOSAGES

- **CKD (oral):** 0.25 mcg daily or alternate days, adjusted to calcium/PTH (usual 0.5–1 mcg/day).
- **Hypoparathyroidism:** 0.25–2 mcg daily.
- **Osteoporosis:** 0.25 mcg twice daily.

PHARMACOKINETICS

- **Onset:** 2–6 h. **Duration:** 3–5 days.
- Already active — bypasses renal 1α-hydroxylation.
- Metabolised in liver/kidney; excreted in bile. **t½:** 5–8 h.

PHARMACODYNAMICS

- Acts on **vitamin D receptors** → ↑ intestinal calcium and phosphate absorption, ↑ bone mineralisation, ↓ PTH secretion.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Usually none at correct dose.
- Early hypercalcaemia: nausea, constipation, weakness, thirst, polyuria.

ADVERSE EFFECTS (SERIOUS)

- Hypercalcaemia and hypercalciuria → kidney stones, nephrocalcinosis, arrhythmias.
- Vascular/soft-tissue calcification (high Ca × P product).

CONTRAINDICATIONS & CAUTIONS

- Hypercalcaemia; vitamin D toxicity.
- Metastatic calcification.

KEY INTERACTIONS

- **Thiazides:** ↑ hypercalcaemia.
- **Digoxin:** arrhythmias with hypercalcaemia.
- **Magnesium antacids:** hypermagnesaemia in renal failure.
- **Cholestyramine:** ↓ absorption.

NURSING MANAGEMENT

- Monitor serum calcium weekly initially, phosphate, PTH, creatinine.
- Watch for symptoms of hypercalcaemia.
- Dietary calcium review.

PATIENT EDUCATION

- Report nausea, constipation, excessive thirst, confusion or weakness.
- Do not take other vitamin D supplements unless prescribed.
- Keep blood test appointments.

KEY POINTS

- Active vitamin D — no kidney activation needed.
- Hypercalcaemia; monitor Ca closely.
- CKD and hypoparathyroidism.
- Microgram doses (0.25 mcg).